Executive Summary

Customer churn presents a significant challenge for telecommunications companies because customer attrition can reduce recurring revenue and increase the cost of acquiring replacement customers. This project applies machine learning to the Cell2Cell telecommunications customer dataset to develop and evaluate an artificial neural network (ANN) for identifying customers at risk of churning. The objective is to support more proactive, data-driven retention strategies by identifying high-risk customers and informing the prioritization of marketing and customer retention campaigns.

The analysis used the Cell2Cell training dataset, comprising 51,047 customer records and 58 original columns. Initial data preparation involved examining missing values, checking for duplicate records, assessing the distribution of the churn target, and preparing features for modelling. Customer identifiers were excluded from the predictors, while retention-related variables that could introduce target leakage were excluded from the primary model. Missing numerical and categorical values were handled through imputation, numerical features were standardized, and categorical features were one-hot encoded. The dataset was divided into stratified training, validation, and test subsets to preserve the distribution of churn and non-churn customers. Preprocessing transformations were fitted using the training data to reduce the risk of data leakage.

The baseline ANN consisted of two hidden layers containing 32 and 16 neurons, respectively, with ReLU activation, followed by a sigmoid output layer for binary churn classification. The model was trained using the Adam optimiser and binary cross-entropy loss, with early stopping to limit overfitting. At the default classification threshold of 0.50, the ANN achieved a test accuracy of 70.68%, precision of 42.96%, recall of 5.26%, F1-score of 0.0937, and ROC-AUC of 0.6314. Although the accuracy appeared reasonable, the low recall indicated that the model failed to identify most customers who actually churned. This demonstrated why accuracy alone is insufficient for assessing a churn prediction model.

To address this limitation, the classification threshold was selected using validation data by maximising the F1-score. The ANN's selected threshold was 0.23. At this threshold, test recall increased to 85.59% and the F1-score improved to 0.4783, while precision was 33.19% and accuracy was 46.19%. The model correctly identified 1,889 of the 2,207 actual churners in the test set, although it also incorrectly flagged 3,803 non-churning customers. A logistic regression benchmark was evaluated using the same data split and general preprocessing approach. Its validation-selected threshold was 0.22, producing test recall of 83.78%, precision of 32.47%, F1-score of 0.4680, and ROC-AUC of 0.6128. At their respective validation-selected thresholds, the ANN performed better than logistic regression across the reported test metrics, although the differences were modest.

These findings demonstrate that classification-threshold selection can substantially improve churn detection and that the ANN provides a promising starting point for customer retention analysis. However, its false-positive rate and moderate ranking performance mean that further evaluation is required before operational deployment. Recommended next steps include

assessing Precision@10% and Lift@10% to measure how effectively each model prioritises customers within a limited campaign budget, testing additional model optimisation strategies, and evaluating the commercial costs and benefits of retention interventions. Overall, the project illustrates how machine learning can support telecommunications customer retention while highlighting the importance of leakage prevention, appropriate evaluation metrics, threshold calibration, and business-focused model assessment.